

PROJECT TESTING

POCKETSMART AI — Smart Budget & Recommendation Assistant

Expanded phase evidence with detailed wording, diagrams, workflow, and project-focused explanation.

Testing objective

Testing verifies that the documented PocketSmart AI workflow behaves as expected from authentication through recommendation generation and History. Testing is not limited to whether a page opens; it checks whether inputs are accepted, protected access works, recommendations are generated, budget summaries appear, and History can display previous records.

Functional verification

Registration and login should allow valid users to authenticate. Protected pages should require authentication and logout should end the session. Home, Party, and Jewelry requirements should be accepted by their respective planners. The selected planner should generate a structured recommendation, and the result should display total and remaining budget information where applicable.

Security verification

The project documentation identifies several security considerations: passwords are hashed before storage; JWT/access tokens are stored through HTTP-only cookies; the Gemini API key is protected through environment configuration; planner inputs and uploaded images are validated; and History records are associated with the relevant authenticated user. These checks help connect visible behaviour with responsible backend handling.

Visual Model



Figure — Phase visual model

Process Flow



Figure — Detailed workflow / roadmap

PHASE 6 — DETAILED CONTENT

PocketSmart AI connects practical planning with a structured recommendation workflow. The following section expands the phase in clear project language so the document has sufficient explanatory depth for academic/project review.

Project Explanation

A strong budget-planning assistant should reduce uncertainty without hiding the reasoning context behind the result. In PocketSmart AI, the user supplies the information that matters for the selected category, including budget and relevant preferences. The application then processes that context and presents a structured recommendation. This structure makes the output easier to read, compare, and explain during a project demonstration. The project also demonstrates how a common architecture can support different use cases. Home Interior planning focuses on rooms, furniture, lighting, fans, and dining requirements. Party planning focuses on guests, event type, venue type, catering, decoration, and entertainment. Jewelry planning focuses on occasion, style preferences, budget, and optional outfit-image context. Although the inputs differ, the overall application pattern remains consistent.

Key Concepts

Test Area	Expected behaviour
Registration / Login	Valid users can create accounts and authenticate.
Protected Access / Logout	Protected pages require authentication; logout ends the session.
Planner Inputs	Home, Party, and Jewelry requirements are accepted.
Recommendation Generation	Selected planner generates structured recommendations.
Budget Summary	Total and remaining budget are displayed.
History	Previous recommendation records can be viewed.

Roadmap / Mind-Map View

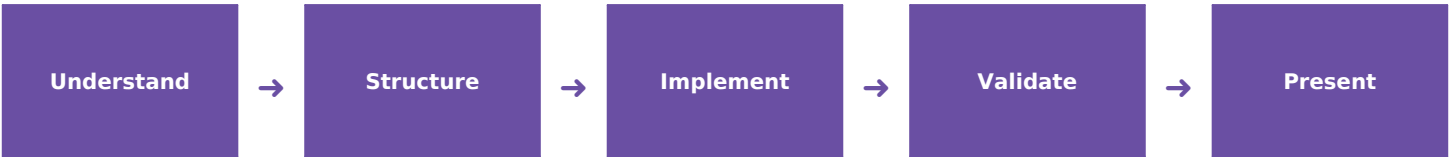


Figure — Additional project roadmap visual

Project Evidence

The supplied project documentation includes interface and workflow evidence. A representative screenshot is included here to connect the written explanation with the implemented application.

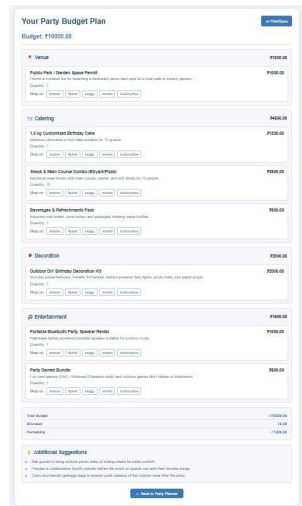
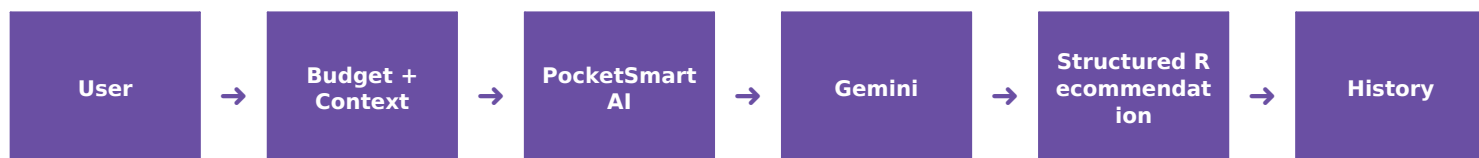


Figure — Representative PocketSmart AI project evidence

PHASE SUMMARY

Phase 6 contributes to the overall PocketSmart AI lifecycle by making the project easier to understand, build, validate, or present. The emphasis in this phase is on clear communication: the reader should be able to see what the system is doing, why the step matters, and how it connects to the final budget-aware recommendation workflow.

End-to-End Connection



Project Reflection

PocketSmart AI is designed around a practical user problem: making planning decisions while respecting a defined budget. The project brings together web development, authentication, structured data, Generative AI, visual presentation, and History. This combination makes the application a useful example of how a software project can move from a simple idea to a demonstrable end-to-end system. The documentation should therefore communicate not only the features, but also the reasoning, workflow, and evidence behind them.

Closing Note

The phase is complete when its purpose, workflow, supporting evidence, and connection to PocketSmart AI are understandable to a reviewer without requiring them to infer missing steps. The expanded paragraphs and diagrams in this version are intended to make that explanation clearer and more professional.